

Assignment No -2

Assignment Title

Prompt Engineering Techniques

Problem Statement

Design and implement a system to evaluate different prompt engineering techniques including:

- Zero-Shot Prompting
- Few-Shot Prompting
- Chain-of-Thought (CoT) Prompting

Apply these techniques to solve the same problem and compare their:

- Quality
- Accuracy
- Reasoning Ability

Step 1: What is Prompt Engineering?

Imagine you have a very intelligent assistant.

The answer you get depends on **how you ask the question**.

The question you ask to an LLM is called a **Prompt**.

Example:

Prompt:

What is AI?

Response:

Artificial Intelligence is...

Another Prompt:

Explain AI to a 10-year-old using an example.

Response:

Imagine a robot learning like a child...

Same AI.

Different prompt.

Different answer.

This art of writing better prompts is called

Prompt Engineering

Think of it like asking your teacher questions.

Poor Question

Explain.

Good Question

Explain Linear Regression with a house price example, mathematical formula, Python code, and diagram.

The second prompt gets a much better answer.

Step 2: What are Prompt Engineering Techniques?

There are many techniques.

Your assignment covers three.

1. Zero-Shot Prompting
2. Few-Shot Prompting
3. Chain-of-Thought Prompting

Let's understand one by one.

Step 3: Zero-Shot Prompting

Zero means

"No examples are given."

You directly ask the AI.

Example

Prompt

Translate this sentence into Marathi.

I love Artificial Intelligence.

AI replies.

No example is provided.

This is Zero-Shot.

Another Example

Summarize this paragraph.

No sample summary is shown.

AI directly performs the task.

Diagram

User --→ "What is Machine Learning?" ----→ LLM ---→ Answer

No examples.

No hints.

Only instruction.

Step 4: Few-Shot Prompting

Few means

"Give some examples first."

Example

Prompt

English : Marathi

Dog -> कुत्रा

Cat -> मांजर

Horse -> ?

The AI understands the pattern.

It answers

घोडा

Here we taught the AI using examples.

Another Example

Positive Review

"I loved this movie."

Negative Review

"This movie was boring."

Now classify:

"The acting was excellent."

AI learns from examples.

Diagram

Examples --> Dog → कुत्रा ---> Cat → मांजर --> New Question --> Horse → ? ->

Answer -> घोडा

Step 5: Chain-of-Thought Prompting

This is the most interesting.

Instead of asking

What is 25×16 ?

You ask

Solve step by step.

AI starts reasoning.

Example

$$25 \times 16$$

$$25 \times 10 = 250$$

$$25 \times 6 = 150$$

$$250 + 150 = 400$$

Answer

400

Another Example

Question

A shopkeeper bought a pen for ₹20 and sold it for ₹35.

Find profit.

Normal Prompt

Find profit.

Chain of Thought Prompt

Think step by step.

Find the Cost Price.

Find the Selling Price.

Use Profit = SP – CP.

Then give the answer.

The reasoning becomes visible.

Diagram

Question -> Reasoning -> Intermediate Steps --> Final Answer

Step 6: What will your assignment do?

You will ask

The SAME question

using

Technique 1

Technique 2

Technique 3

Then compare the outputs.

Example Question

Explain Linear Regression.

Technique 1

Zero Shot

Explain Linear Regression.

Technique 2

Few Shot

Example 1

Decision Tree

A model that...

Example 2

Logistic Regression

A model that...

Now explain Linear Regression.

Technique 3

Chain of Thought

Explain Linear Regression.

Think step by step.

Start with definition.

Then formula.

Then example.

Then Python implementation.

Finally applications.

Notice how the prompt guides the model through a structured reasoning process.

Step 7: What software do you need?

You already have

✓ VS Code

Need

Python

Google Gemini API

.env file

Folder

PromptEngineering

|

app.py

.env

requirements.txt

Step 8: Install Libraries

Open Terminal

```
pip install google-genai
pip install python-dotenv
```

Step 9: Create .env

```
GEMINI_API_KEY=YOUR_API_KEY
```

Step 10: Write Python Code

```
import os
from dotenv import load_dotenv
from google import genai

load_dotenv()

client = genai.Client(
    api_key=os.getenv("GEMINI_API_KEY")
)
```

Now your AI is connected.

Step 11: Zero-Shot Prompt

```
prompt = """
Explain Artificial Intelligence.
"""

response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents=prompt
)
```

```
print(response.text)
```

Step 12: Few-Shot Prompt

```
prompt = """
```

Example:

Question:

What is Python?

Answer:

Python is a programming language.

Example:

Question:

What is Java?

Answer:

Java is also a programming language.

Now answer:

Question:

What is Artificial Intelligence?

```
"""
```

Run.

Observe the output.

Step 13: Chain-of-Thought Prompt

```
prompt = """
```

Explain Artificial Intelligence.

Think step by step.

1. Define AI

2. Explain how it works

3. Give one real-life example

4. Explain advantages

5. Explain disadvantages

6. Conclude

""""

Run again.

Step 14: Compare Results

Create a table like this:

Technique	Prompt Style	Response Quality	Accuracy	Reasoning
Zero-Shot	Direct instruction	Good	High	Low
Few-Shot	Examples provided	Better	High	Medium
Chain-of-Thought	Step-by-step guidance	Best	High	High

Step 15: Observe the Differences

Zero-Shot

- Fast
- Simple
- No examples
- Good for straightforward tasks

Few-Shot

- Learns from examples
- Better formatting
- Better consistency

Chain-of-Thought

- Shows reasoning
- Better for mathematics
- Better for logical problems
- Better for complex tasks

Step 16: Example Assignment Output

Suppose the question is:

What is Machine Learning?

Zero-Shot

Machine Learning is a subset of AI...

Short answer.

Few-Shot

AI

Definition...

Deep Learning

Definition...

Machine Learning

Definition...

Better structure.

Chain-of-Thought

Step 1

AI means...

Step 2

Machine Learning is part of AI...

Step 3

Algorithms learn from data...

Step 4

Example...

Step 5

Conclusion...

Very detailed reasoning.

Step 17: What should you submit?

Your assignment should include:

1. **Objective** – Explain that the goal is to compare prompt engineering techniques.
2. **Theory** – Briefly describe Prompt Engineering, Zero-Shot, Few-Shot, and Chain-of-Thought prompting.
3. **Software Requirements** – Python, VS Code, google-genai, python-dotenv.
4. **Implementation** – Python code that sends three different prompts to the same LLM.
5. **Screenshots** – Terminal outputs for each prompting technique.
6. **Comparison Table** – Compare response quality, accuracy, and reasoning.
7. **Conclusion** – Summarize which prompting technique worked best for the chosen problem and why.

Workflow Summary



